Omjee R Giri

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Professional Summary

AI/ML Engineer and Quant Researcher (BIT Mesra + IIT Madras) with production experience in LLM orchestration, RAG pipelines, vector databases, graph neural networks, and model deployment. Codeforces Specialist; GSSoC 2026 Global Rank #1,065 with 7 merged PRs to a live platform. Targeting AI/ML Engineering, Software Engineering, Data Science, and Quant Research internships.

Education

Birla Institute of Technology, Mesra Aug 2024 – Present | GPA: 7.6 / 10
Bachelor of Technology – Artificial Intelligence and Machine Learning

Indian Institute of Technology, Madras Jan 2025 – Present | GPA: 7.0 / 10
Bachelor of Science – Data Science and Applications

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Linear Algebra, Probability & Statistics, DBMS, Computer Networks, Operating Systems, OOP

Work Experience

Birla Institute of Technology, Mesra — Dept. of Computer Science Jan 2026 – Present
Undergraduate Research Assistant — Legal Lens: AI Legal Document Intelligence

- Engineered FastAPI data pipelines for 100+ legal PDFs (ingestion, parsing, chunking, indexing, deployment) with sub-200ms latency
- Designed RAG pipelines with metadata tagging, cutting irrelevant retrieval by 40%; evaluated via precision@k, mean reciprocal rank, and latency scoring
- Implemented vectorless Page Indexing in place of ANN vector retrieval, reducing index build complexity from $O(n \log n)$ to $O(n)$ and build time by 60%
- Architected a 4-layer system (API routing, document processing, AI services, database) for independent scaling, reducing coupling by ∼50%

Tech Stack: Python, FastAPI, RAG Pipelines, NLP, REST APIs, Feature Engineering, model deployment, Git

GirlScript Summer of Code (GSSoC) 2026 May 2026 – Present
Open Source Software Engineer Intern — Global Rank #1,065

- Engineered legal-citation validation across 2 Indian statutes (BNS, BNSS) — added 10+ range rules and 40+ Pytest cases, reducing errors by ∼70%
- Built fault-tolerant LLM orchestration across 3+ providers with exponential-backoff retry, cascading fallback, and monitoring (error rates, p95 latency), raising uptime to ∼99.5%; delivered token-bucket rate limiting and service-layer modularization across 7 PRs with zero regressions

Tech Stack: Python, FastAPI, Pytest, REST APIs, CI/CD, Docker, GitHub Actions | 7 PRs merged | 5 issues raised

Projects

Legal Lens: AI Legal Document Intelligence GitHub | Flagship Research Project
Hybrid semantic + knowledge-graph retrieval over Indian statutory law — developed as Undergraduate Research Assistant, BIT Mesra CS Dept.

- Engineered an end-to-end RAG pipeline — ingestion, semantic chunking, dense embedding generation, cosine-similarity retrieval over L2-normalized 384-dim vectors, prompt construction, and Groq LLM generation; added in-memory caching to eliminate redundant retrieval
- Implemented hybrid retrieval fusing embedding-based semantic search with Neo4j Cypher traversal of a directed property graph (statute nodes, section relationships), unifying both tracks into one LLM context payload with fallback on empty graph results
- Architected a modular FastAPI backend (API, Services, Schemas, Core layers) with versioned routing, Pydantic schema validation, and end-to-end validation across 11 development phases
- Benchmarked retrieval quality via precision@k and mean reciprocal rank over a 100+ document statutory corpus at sub-200ms query latency; containerised with Docker for reproducible deployment

Tech Stack: Python, FastAPI, Groq LLM, Neo4j, Sentence Embeddings, RAG Pipelines, Pydantic, NLP, Docker

Agentic Precision-Psychiatry System on Multi-Branch qEEG Subtyping GitHub | Research Project

- Building a multi-branch hybrid classifier (1D-CNN + Graph Convolutional Network + Transformer) for 6-class qEEG subtyping from 19-channel EEG; the GCN branch propagates over a symmetric-normalized adjacency $D^{-1/2}AD^{-1/2}$ from inter-channel coherence, with multi-head self-attention over temporal dependencies
- Implemented the signal pipeline (band-pass filtering, ICA artifact removal, re-referencing, epoching) with spectral (Welch PSD band power, frontal α-asymmetry), connectivity (phase-locking value, coherence), and nonlinear (sample entropy, Hjorth parameters) features
- Orchestrating acquisition-to-recommendation in n8n (file-drop → preprocessing → features → inference → agent); Recommendation Agent maps predicted subtype to structured pharmacotherapy / TMS-target / psychotherapy output and auto-generates a clinician report

Tech Stack: Python, PyTorch, 1D-CNN, GCN, Transformer, MNE-Python, n8n, NumPy, SciPy, FastAPI, RAG, Docker | Target: IEEE JBHI

ARGUS: Multi-Agent Quantitative Research Terminal GitHub | Personal Project

- Architected a 6-layer FastAPI backend (Market Data, Quant Analytics, Portfolio Analytics, Multi-Agent AI, RAG Knowledge, API Services) with async services and dependency injection
- Built a deterministic Quant Engine computing annualized Sharpe $\left(E[R_p - R_f]/\sigma_p\right)\sqrt{252}$, CAPM Beta via $\text{Cov}(R_i, R_m)/\text{Var}(R_m)$, max drawdown and rolling volatility from live yfinance data; integrated ChromaDB as a RAG Knowledge Layer with provenance-aware citations over 10-Ks, 10-Qs, and earnings calls
- Implemented a Portfolio Analytics Engine for concurrent multi-asset retrieval and covariance-matrix risk decomposition; 56+ automated tests across analytics, retrieval, and agents

Tech Stack: Python, FastAPI, AsyncIO, ChromaDB, Pydantic, yfinance, Multi-Agent Systems, RAG, Docker, WebSockets, TensorFlow

Amazon-Style Semantic Product Recommendation Engine GitHub | Personal Project

- Built a semantic recommendation system on all-MiniLM-L6-v2 384-dim embeddings with FAISS IndexFlatL2 exact nearest-neighbour search ($O(nd)$ per query), achieving sub-second retrieval across 50,000+ product vectors and ∼85% latency reduction vs keyword search
- Designed a 5-signal weighted ranking function over min-max normalized features (similarity 0.35, ratings 0.25, popularity 0.15, budget 0.10, quality 0.05), improving relevance by ∼35% over the keyword baseline

Tech Stack: Python, Sentence Transformers, FAISS, Hugging Face, Pandas, NumPy, Streamlit, scikit-learn, Keras, Unittest

Achievements

- **GSSoC 2026** — Global Rank #1,065 among 70,000+ registered contributors; 7 PRs merged into a production codebase
- **Codeforces** — Specialist (max rating 1400+); active rated-contest participant

Technical Skills

Languages: Python, C++, C, TypeScript, SQL, HTML/CSS
ML and AI: PyTorch, TensorFlow, Keras, scikit-learn, Sentence Transformers, Hugging Face, FAISS, GNNs, Transformers, NLP, Deep Learning, LLMs, MLflow
Data and Quant: NumPy, Pandas, SciPy, Matplotlib, Seaborn, ARIMA, Time Series, Statistical Modeling, Backtesting, Monte Carlo, Sharpe Ratio, VaR, RSI, MACD
Backend and DB: FastAPI, Flask, REST APIs, WebSockets, RAG Pipelines, SQLAlchemy, Redis, Pydantic, MySQL, PostgreSQL, Neo4j, ChromaDB, Vector Databases
DevOps / Tools: Docker, GitHub Actions, CI/CD, Git, Linux, Pytest, Unittest, Postman, Swagger/OpenAPI, Streamlit, RAG Evaluation, Data Engineering
Core CS: DSA, Algorithms, System Design, OOP, Computer Networks, DBMS, OS, Concurrent Programming